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Mental Health Analyzer - User Manual

Table of Contents

1. Introduction
2. Getting Started
3. Using the Application
4. Understanding Results
5. Monitoring Dashboard
6. Troubleshooting
7. FAQ
8. Support and Contact

Introduction

The Mental Health Analyzer is an application designed to help analyze journal entries or text statements and categorize them into different mental health states. By using machine learning technology, the system can identify patterns associated with various mental health conditions, providing insight into emotional and psychological states expressed in text.

Purpose and Benefits

This application serves as a tool for:

- Self-reflection and awareness of mental health states
- Identifying potential areas of concern in emotional well-being
- Tracking mental health patterns over time
- Providing supportive guidance based on identified emotional states

IMPORTANT DISCLAIMER: This system is not a diagnostic tool and should not replace professional mental health evaluation or treatment. The categorizations provided are based on statistical patterns and should be considered as informational only.

If you or someone you know is experiencing a mental health crisis, please contact a mental health professional or crisis service immediately.

Getting Started

System Requirements

- Docker and Docker Compose installed
- Web browser (Chrome, Firefox, Safari, or Edge)
- Minimum 4GB RAM and 2 CPU cores recommended
- Internet connection for initial setup

Installation

1. Clone or download the application:

```
git clone <https://github.com/><repo>/mental-health-analysis.git  
cd mental-health-analysis
```

2. Build and start the containers:

```
docker-compose build  
docker-compose up -d
```

3. Start Prometheus monitoring

In a separate terminal:

```
prometheus --config.file=./prometheus/prometheus.yml
```

4. Verify services are running

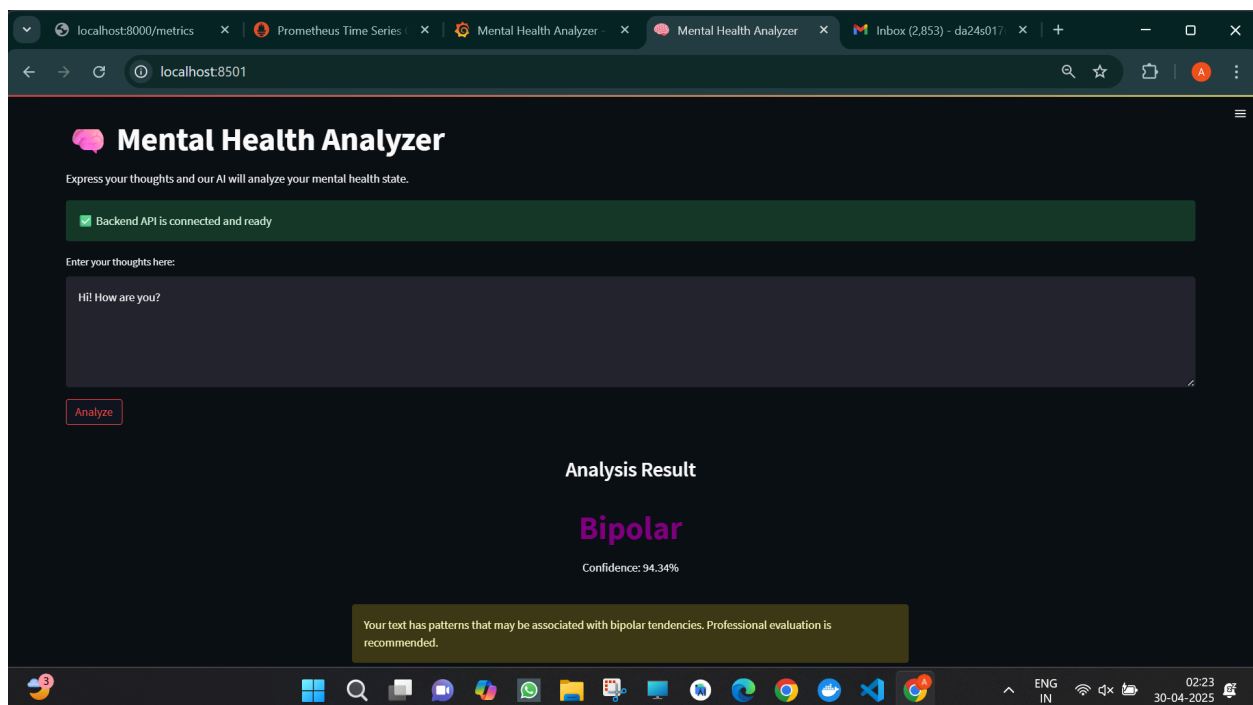
- Frontend UI: <http://localhost:8501>
- API Backend: <http://localhost:8000>
- API Backend metrics: <http://localhost:8000/metrics>
- Prometheus: <http://localhost:9090>

- Grafana: <http://localhost:3000>

5. Configure Grafana (first-time setup)

- Log in to Grafana (<http://localhost:3000>) using default credentials:
 - Username: admin
 - Password: admin
- Add Prometheus as a data source:
 - Navigate to Configuration > Data Sources
 - Click "Add data source" and select "Prometheus"
 - Set URL to <http://localhost:9090>
 - Click "Save & Test"
- Navigate to Dashboard - Mental Health Analyzer to view the metrics.

Using the Application



Accessing the Application

1. Open your web browser and navigate to <http://localhost:8501>
2. You should see the Mental Health Analyzer interface with a text input area

Analyzing Journal Entries

1. Enter your journal entry or statement

- Type or paste your text in the provided text box
- For best results, write at least a few sentences that express your thoughts or feelings

2. Submit for analysis

- Click the "Analyze" button below the text input area
- Wait a few moments for the system to process your entry

3. View results

- The analysis result will appear below the input area
- You'll see the detected mental health category and a supportive message

Best Practices for Journal Entries

- Be honest and authentic in your writing
- Focus on expressing your thoughts and emotions
- Write in your natural style without trying to influence the analysis
- Include context about situations or experiences when relevant
- Aim for at least 3-4 sentences to provide sufficient information for analysis

Understanding Results

Mental Health Categories

The system can identify the following categories:

Category	Description
Personality Disorder	Indicates inflexible and maladaptive patterns of thinking, feeling and behaving.
Depression	Indicates patterns associated with depressive thoughts or feelings

Anxiety	Reflects worry, fear, or anxious thought patterns
Stress	Shows signs of being under pressure or experiencing strain
Bipolar	Contains elements of mood fluctuation or extremes
Normal	Shows balanced emotional patterns without significant distress
Suicidal	Shows suicidal tendencies. Needs immediate medical attention.

Interpreting Results

- **Category:** The primary mental health state detected in your text
- **Confidence:** How strongly the pattern matches the identified category
- **Supportive Message:** Guidance or encouragement based on your result

Important Considerations

- Results are based on language patterns only and are not diagnostic
- A single journal entry provides limited information - patterns over time are more meaningful
- External factors like writing style can influence results
- Always consult a mental health professional for concerns about your well-being

Monitoring Dashboard



Accessing Grafana Dashboard

1. Open your web browser and navigate to <http://localhost:3000>
2. Log in using your Grafana credentials
3. Select the Mental Health Analyzer dashboard from the dashboard list

Dashboard Elements

The dashboard displays various metrics about the system:

1. **System Metrics**
 - CPU and memory usage
 - Disk IO Rate
 - Network IO Rate
 - Free Disk space
2. **Application Metrics**
 - Prediction counts by category
 - Category Distribution
 - Count of API call requests

Using Dashboard Data

- Monitor system health and performance
- Track usage patterns
- Understand distribution of mental health categories across entries

Troubleshooting

Common Issues and Solutions

Issue	Possible Cause	Solution
Frontend not loading	Docker container not running	Check with <code>docker-compose ps</code> and restart if needed
"Cannot connect to backend" error	Backend service unavailable	Ensure backend container is running and healthy
Slow response times	System under heavy load	Check system resources and consider scaling if needed
"Model not found" error	Missing model file	Verify model file exists in the specified path
Unexpected category results	Text too short or ambiguous	Provide more detailed and expressive text

Checking Logs

To view application logs:

View frontend logs

`docker-compose logs frontend`

View backend logs

`docker-compose logs backend`

Follow logs in real-time

`docker-compose logs -f`

Restarting Services

If you encounter issues, try restarting the services:

Restart individual service

`docker-compose restart frontend`

```
docker-compose restart backend
```

```
# Restart all services
```

```
docker-compose down
```

```
docker-compose up -d
```